

Medical Electronics: Key Concerns and Opportunities

The future of medical electronics will be paved with technologies that allow portability, connectivity and data security



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Electronic devices are increasingly being seen in the field of medicine for diagnosis, therapy and rehabilitation. Medical electronics and electromechanical equipment have become indispensable for the better service of patients not only in hospitals but also outside the hospitals. The service quality of hospitals is often judged by the state of the equipment they have at their disposal.

Consider a few cases to understand the reach and impact of medical elec-

tronics on patient care. Surgeons use three-dimensional (3D) body registration systems and high-resolution imaging for cutting into invisible areas. Also, implanted insulin pumps with a closed loop-feedback are capable of automatic fine-tuning of drug delivery, thus cutting down the need for diabetics to opt for regular self-tests. Unlike oral psychiatric drugs, implanted stimulators and drug delivery devices focus only on affected parts reducing the risk of collateral damages and side effects.

Today, electronic therapeutic and diagnostic instruments and techniques

are blended innovatively for overall improvement in healthcare and devising potential cure for a large number of hitherto incurable diseases. With growing popularity of medical electronics, there is now a growing concern about quality, safety, reliability, cost, liability and regulatory issues.

Design, repair and maintenance

Medical electronics engineering ranges from model-driven embedded software design to PCB design and manufacture, and a large number of inter-related sub-sectors too. It covers a very wide range of technologies including radio frequency, analogue semiconductors, digital and microprocessor chips, digital signal processors, sensors, actuators, electromagnetics, optoelectronics and photonics, displays, embedded software, power supplies and antennae.

The rapid advancement in information technology and healthcare consciousness has accelerated the scope for medical electronics. Fast growth in medical electronics is further influencing various demographic trends like consumers' expectations of more household medical electronic equipment, enhanced portability of complex imaging and monitoring systems, further miniaturisation of implantable equipment with lower energy consumption, and functional integration of equipment and applications in wireless and network technology.

A large number of hospitals are equipped with high-tech medical equipment but lack trained manpower. This results in long down-time and early classification of equipment as defunct. Routine preventive maintenance is required to ensure proper working condition of